

Weekly CT - 03

Cambridge International AS and A Level Computer Science 9618

Data representation | Networks | Hardware and memory

Time: 30 minutes

Total marks: 30

Instructions: Answer all questions. Show working for calculations. Write clearly in the spaces provided.

1. Run-length encoding (RLE)

[3]

(a) A row of bitmap pixels is: R R R R R B B G G G G G R R R. Encode this row using RLE. You may write each run as count + symbol, for example 3R. [2]

(b) Explain why RLE is a lossless compression method. [1]

2. Image representation and file size

[3]

(a) State what a colour depth of 24 bits means. [1]

(b) An uncompressed bitmap has a resolution of 1280 x 720 pixels and a colour depth of 24 bits. Ignore the file header. Calculate the file size in MiB. Show your working. [2]

Working:

Answer (MiB): _____

3. Sound representation and file size

[4]

(a) State what is meant by sample rate. **[1]**

(b) State what is meant by sample resolution (bit depth). **[1]**

(c) A 10-second mono sound clip is sampled at 22,050 samples per second using 16 bits per sample. Calculate the uncompressed file size in bytes. Show your working. **[2]**

Working:

Answer (bytes): _____

4. Network devices

[5]

Write the correct network device for each description. Use each of the following once: router, switch, modem, gateway, hub.

No.	Description	Device
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No.	Description	Device
1	Forwards packets between different IP networks using destination IP addresses.	_____
2	Connects devices within a LAN and forwards frames to the correct port using MAC addresses.	_____
3	Converts/modulates and demodulates signals so a network can communicate over an ISP access line.	_____
4	Connects networks that use different protocols and can perform protocol translation.	_____
5	Sends incoming data out to all other ports rather than selecting one destination port.	_____

5. Classful and classless IPv4 addressing / subnetting

[5]

(a) Under classful IPv4 addressing, state the address class of 172.16.8.20. [1]

(b) State the default subnet mask for that class. [1]

(c) A device has the classless address 192.168.10.70/26. State the subnet mask. [1]

(d) State the network address for 192.168.10.70/26. [1]

(e) State the broadcast address for 192.168.10.70/26. [1]

6. Cloud computing

[3]

(a) Define cloud computing. [1]

(b) State one benefit of using cloud computing. [1]

(c) State one drawback of using cloud computing. [1]

7. Storage hierarchy, swap space and page faults

[3]

(a) Place the following in order from closest/fastest to the CPU to farthest/slower: cache, RAM, registers, secondary storage (SSD/HDD). [1]

(b) As you move farther from the CPU in the storage hierarchy, what generally happens to cost per bit? [1]

(c) Explain what a page fault is and state how swap space may be involved. **[1]**

8. Types of RAM and ROM

[3]

(a) State one difference between SRAM and DRAM. **[1]**

(b) Name the type of ROM that can be erased using ultraviolet light and then reprogrammed. **[1]**

(c) Name the type of ROM that can be electrically erased and reprogrammed. [1]

9. How long did you study for this short exam? [0.5]

Hour = _____

10. Thank you for participating in the exam honestly. [0.5]